Infant Perceptual Narrowing for Speech Contrasts

During the first year, discrimination of native-language phonetic contrasts is maintained or strengthened while sensitivity to some nonnative contrasts declines; this is attunement, not a permanent closing of language learning.

ADE: HIGH-LAB; MODERATE-TIMETABLE; LOW-SEALED-DEADLINE | STATUS: STABLE | DOMAIN: LANGUAGE | REPLICATION: VERIF

WHAT WE KNOW (Narrowest Supported Conclusion)

Young infants can discriminate phonetic contrasts they have not yet needed for a native language. Across the first year, that broad sensitivity is reorganized by experience: native-language contrasts remain discriminable, while some nonnative contrasts become harder. This is attunement to the languages actually heard, not a deletion of a universal dictionary.

EVIDENCE AND CAUSAL ASSESSMENT

Werker & Tees (1984) compared English infants with English and Salish adults on a Salish (non-English) contrast, and combined cross-sectional with longitudinal infant data. Infants discriminated the nonnative contrast; the decline appeared within the first year and depended on specific language experience. Kuhl, Williams, Lacerda, Stevens & Lindblom (1992), Science, showed in an independent laboratory that by six months American infants' vowel perception had already been altered by English experience — a conceptual replication of experience-dependent phonetic attunement, not a household-vocabulary study and not a universal 6–12-month clock. Identification is experimental (manipulated contrast × age/experience), not a randomized life-course trial. A 2002 reprint of Werker & Tees remains a landmark; later work has questioned how far the original samples generalize. The timetable and sealed-deadline claims are not replicated as universals and remain Moderate and Low.

EFFECT SIZE AND PERSISTENCE

The outcome is contemporaneous discrimination, not later vocabulary. Persistence of the decline into adulthood is well known for some contrasts; recovery and bilingual maintenance show that the system is not sealed.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory phonetic discrimination task, not an observational population phenotype association).

Historically narrow language samples; looking-time methods; infant state. Diversity of languages and communities remains a live critique of treating 6–12 months as a single clock.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean a rigid deadline after which a second language cannot be learned, or that hearing more than one language confuses the infant. Prenatal preference (DeCasper) is a different claim.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary participation in a linguistic community is the relevant exposure. Phonetic-discrimination training as cognitive enhancement is not justified by this card.

VERIFIED SOURCES & DOI IDENTIFIERS

- Werker & Tees (1984), Infant Behavior and Development
- Kuhl et al. (1992), Science 255:606–608